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In [29]: import pandas as pd
import numpy as np
df=pd.read_csv('breast_cancer_clinical3.csv')
print('breast_cancer_clinical3.csv读取成功')
print('-----')
print('df数据结构: ')
print(df.head())
print(df.info())
print(df.describe())
print(df.isnull().sum())
print('-----')
LN0=df[df['Lymph_Node']==0]
print('LN0的数据结构: ')
print(LN0.describe())
print('-----')
LN1=df[df['Lymph_Node']==1]
print('LN1的数据结构: ')
print(LN1.describe())
print('-----')
print('-----连续变量-----')
print('Age的正态分析: ')
from scipy.stats import shapiro
stat,p=shapiro(df['Age'])
print('Age的p值: ',p)
print('因为Age的p值大于0.05, 所以没有理由认为Age不符合正态分布, 因此下一步需要进行')
stat,p=shapiro(LN0['Age'])
print('LN(-)Age的p值: ',p)
stat,p=shapiro(LN1['Age'])
print('LN(+)Age的p值: ',p)
print('Age的直方图: ')
import matplotlib.pyplot as plt
plt.hist(df['Age'],bins=10)
plt.xlabel('Age')
plt.ylabel('Frequency')
plt.show()
print('Age的Q-Q图: ')
import scipy.stats as stats
stats.probplot(df['Age'],dist='norm',plot=plt)
plt.show()
print('-----')
print('Tumor_Size的正态分析: ')
stat,p=shapiro(df['Tumor_Size'])
print('Tumor_Size的p值: ',p)
print('因为Tumor_Size的p值大于0.05, 所以没有理由认为Tumor_Size不符合正态分布, 因此')
stat,p=shapiro(LN0['Tumor_Size'])
print('LN(-)Tumor_Size的p值: ',p)
stat,p=shapiro(LN1['Tumor_Size'])
print('LN(+)Tumor_Size的p值: ',p)
print('Tumor_Size的直方图: ')
plt.hist(df['Tumor_Size'],bins=10)
plt.xlabel('Tumor_Size')
plt.ylabel('Frequency')
plt.show()
print('Tumor_Size的Q-Q图: ')
stats.probplot(df['Tumor_Size'],dist='norm',plot=plt)
plt.show()
print('-----')
print('Ki67的正态分析: ')

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stat,p=shapiro(df['Ki67'])
print('Ki67的p值: ',p)
print('因为Ki67的p值小于0.05, 因此Ki67不符合正态分布, 因此下一步要进行Mann Whitney U检验')
print('Ki67的直方图: ')
plt.hist(df['Ki67'],bins=10)
plt.xlabel('Ki67')
plt.ylabel('Frequency')
plt.show()
print('Ki67的Q-Q图: ')
stats.probplot(df['Ki67'],dist='norm',plot=plt)
plt.show()
print('-----')
from scipy.stats import ttest_ind
print('Age的t检验: ')
t,p=ttest_ind(LN0['Age'],LN1['Age'],equal_var=False)
print('Age_ttest的p值: ',p)
print('因为Age_ttest的p值小于0.05, 所以两组的年龄差异有统计学意义。')
print('-----')
print('Tumor_Size的t检验: ')
t,p=ttest_ind(LN0['Tumor_Size'],LN1['Tumor_Size'],equal_var=False)
print('Tumor_Size_ttest的p值: ',p)
print('因为Tumor_Size的p值小于0.05, 所以两组的肿瘤大小差异有统计学意义。')
print('-----')
print('Ki67的Mann Whitney U检验: ')
from scipy.stats import mannwhitneyu
u,p=mannwhitneyu(LN0['Ki67'],LN1['Ki67'],alternative='two-sided')
print('Ki67_U检验的p值: ',p)
print('因为Ki67的p值大于0.05, 所以两组的Ki67差异有统计学意义。')
print('-----')
print('-----分类变量-----')
table1=pd.crosstab(df['ER'],df['Lymph_Node'],normalize=True)*100
print('ER与Lymph_Node的交叉表: ',table1)
table2=pd.crosstab(df['PR'],df['Lymph_Node'],normalize=True)*100
print('PR与Lymph_Node的交叉表: ',table2)
table3=pd.crosstab(df['HER2'],df['Lymph_Node'],normalize=True)*100
print('HER2与Lymph_Node的交叉表: ',table3)
print('-----')
from scipy.stats import chi2_contingency
print('ER的chi square检验: ')
chi,p,dof,expected=chi2_contingency(table1)
print('ER的p值: ',p)
print('ER的expected:',expected)
print('因为样本量太小, 因此再进行一次fisher检验。')
print('ER的fisher检验: ')
from scipy.stats import fisher_exact
oddsratio,p=fisher_exact(table1)
print('ER的p值: ',p)
print('因为p值大于0.05, 所以ER和淋巴结转移之间没有统计学关联。')
print('-----')
print('PR的chi square检验: ')
chi,p,dof,expected=chi2_contingency(table2)
print('PR的p值: ',p)
print('PR的expected:',expected)
print('因为样本量太小, 因此再进行一次fisher检验。')
print('PR的fisher检测: ')
oddsratio,p=fisher_exact(table2)
print('PR的p值: ',p)
print('因为p值小于0.05, 所以PR与淋巴结转移之间有统计学关联。')
print('-----')
print('HER2的chi square:')

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chi,p,dof,expected=chi2_contingency(table3)
print('HER2的p值: ',p)
print('HER2的expected:',expected)
print('因为样本量太小, 因此再进行一次fisher检验。')
print('HER2的fisher检测: ')
oddsratio,p=fisher_exact(table3)
print('HER2的p值: ',p)
print('因为p值大于0.05, 所以HER2和淋巴结转移有统计学关联。')
print('-----')
print('-----Table1 Baseline Characteristics-----')
Table1=pd.DataFrame({
    'Variable':['Age','Tumor_Size','Ki67','ER','PR','HER2'],
    'Overall':['51.95±8.67','3.20±1.02','30(15,70)','20','20','20'],
    'LN(-)':['45.67±7.33','2.42±0.78','20(15,40)','45%','45%','45%'],
    'LN(+)':['57.09±5.92','3.83±0.72','45.45±31.89','55%','55%','55%']
})
print(Table1)
Table1.to_excel('Baseline Characteristics Table1.xlsx')
print('Baseline Characteristics Table1.xlsx保存成功')
print('-----')
print('-----Univariate Logistic Regression Analysis-----')
import statsmodels.api as sm
uni_results=[]
variables=['Age','Tumor_Size','Ki67','ER','PR','HER2']
for variable in variables:
    y=df['Lymph_Node']
    X=df[[variable]]
    X=sm.add_constant(X)
    model=sm.Logit(y,X).fit(dis=False)
    beta=model.params[variable]
    OR=np.exp(model.params[variable])
    CI=np.exp(model.conf_int().loc[variable])
    p=model.pvalues[variable]
    uni_results.append([
        variable,
        OR,
        CI[0],
        CI[1],
        p
    ])
Uni_table=pd.DataFrame(
    uni_results,
    columns=[
        'Variables',
        'Uni_OR',
        'Uni_95%CI_low',
        'Uni_95%CI_high',
        'Uni_P'
    ])
print('Univariate Logistic Regression Analysis Table:')
print(Uni_table)
print('-----')
print('-----Multivariate Logistic Regression Analysis-----')
multi_results=[]
y=df['Lymph_Node']
X=df[['Age','Tumor_Size','Ki67','ER','PR','HER2']]
X=sm.add_constant(X)
model=sm.Logit(y,X).fit(dis=False)
for variable in variables:
    OR=np.exp(model.params[variable])

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CI=np.exp(model.conf_int().loc[variable])
p=model.pvalues[variable]
multi_results.append([
    variable,
    OR,
    CI[0],
    CI[1],
    p
])
Multi_table=pd.DataFrame(
    multi_results,
    columns=[
        'Variables',
        'Multi_OR',
        'Multi_95%CI_low',
        'Multi_95%CI_high',
        'Multi_P'
    ])
print('Multivariate Logistic Regression Analysis Table:')
print(Multi_table)
print('-----')
final_table=pd.merge(Uni_table,Multi_table,on='Variables')
print('-----final_table-----')
final_table['Uni_OR_CI']=(final_table['Uni_OR'].round(2).astype(str)
    +
    '('
    +
    final_table['Uni_95%CI_low'].round(2).astype(str)
    +
    '-'
    +
    final_table['Uni_95%CI_high'].round(2).astype(str)
    +
    ')')
final_table['Multi_OR_CI']=(final_table['Multi_OR'].round(2).astype(str)
    +
    '('
    +
    final_table['Multi_95%CI_low'].round(2).astype(str)
    +
    '-'
    +
    final_table['Multi_95%CI_high'].round(2).astype(str)
    +
    ')')
print(final_table)
print('-----')
print('-----paper table-----')
paper_table=final_table[['Variables','Uni_OR_CI','Uni_P','Multi_OR_CI','Multi_P']
print(paper_table)
paper_table.to_excel('paper_table.xlsx')
print('paper_table.xlsx保存成功')

```

```
breast_cancer_clinical3.csv读取成功
-----
df数据结构:
  Patient_ID  Age  Tumor_Size  Ki67  ER  PR  HER2  Lymph_Node  TNM_Stage  \
0      P001   45         2.1   10   1   1    1           0          II
1      P002   52         3.5   15   1   1    1           1          III
2      P003   38         1.8   20   1   0    0           0           I
3      P004   61         4.2   30   0   0    1           0          III
4      P005   49         2.9   40   1   1    0           1          II

  Molecular_Subtype
0      Luminal_A
1      Luminal_B
2      Luminal_A
3  HER2_positive
4      Luminal_A
<class 'pandas.DataFrame'>
RangeIndex: 20 entries, 0 to 19
Data columns (total 10 columns):
#   Column                Non-Null Count  Dtype
---  -
0   Patient_ID            20 non-null    str
1   Age                   20 non-null    int64
2   Tumor_Size            20 non-null    float64
3   Ki67                  20 non-null    int64
4   ER                    20 non-null    int64
5   PR                    20 non-null    int64
6   HER2                  20 non-null    int64
7   Lymph_Node            20 non-null    int64
8   TNM_Stage             20 non-null    str
9   Molecular_Subtype     20 non-null    str
dtypes: float64(1), int64(6), str(3)
memory usage: 1.7 KB
None

      Age  Tumor_Size      Ki67      ER      PR      HER2  \
count  20.000000   20.00000  20.000000  20.00000  20.000000  20.000000
mean   51.950000    3.19500  41.250000   0.65000   0.450000   0.500000
std     8.666177    1.02108  31.450671   0.48936   0.510418   0.512989
min    36.000000    1.50000  10.000000   0.00000   0.000000   0.000000
25%    46.500000    2.45000  15.000000   0.00000   0.000000   0.000000
50%    51.500000    3.10000  30.000000   1.00000   0.000000   0.500000
75%    58.250000    3.85000  70.000000   1.00000   1.000000   1.000000
max    67.000000    5.10000  95.000000   1.00000   1.000000   1.000000

      Lymph_Node
count  20.000000
mean    0.550000
std     0.510418
min     0.000000
25%     0.000000
50%     1.000000
75%     1.000000
max     1.000000
Patient_ID      0
Age              0
Tumor_Size      0
Ki67            0
ER              0
PR              0
HER2            0
```

Lymph\_Node 0  
TNM\_Stage 0  
Molecular\_Subtype 0  
dtype: int64

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LN0的数据结构:

|       | Age       | Tumor_Size | Ki67      | ER       | PR       | HER2     | \ |
|-------|-----------|------------|-----------|----------|----------|----------|---|
| count | 9.000000  | 9.000000   | 9.000000  | 9.000000 | 9.000000 | 9.000000 |   |
| mean  | 45.666667 | 2.422222   | 36.111111 | 0.888889 | 0.777778 | 0.222222 |   |
| std   | 7.331439  | 0.780669   | 31.993923 | 0.333333 | 0.440959 | 0.440959 |   |
| min   | 36.000000 | 1.500000   | 10.000000 | 0.000000 | 0.000000 | 0.000000 |   |
| 25%   | 42.000000 | 2.000000   | 15.000000 | 1.000000 | 1.000000 | 0.000000 |   |
| 50%   | 45.000000 | 2.300000   | 20.000000 | 1.000000 | 1.000000 | 0.000000 |   |
| 75%   | 48.000000 | 2.600000   | 40.000000 | 1.000000 | 1.000000 | 0.000000 |   |
| max   | 61.000000 | 4.200000   | 90.000000 | 1.000000 | 1.000000 | 1.000000 |   |

| Lymph_Node |     |
|------------|-----|
| count      | 9.0 |
| mean       | 0.0 |
| std        | 0.0 |
| min        | 0.0 |
| 25%        | 0.0 |
| 50%        | 0.0 |
| 75%        | 0.0 |
| max        | 0.0 |

-----  
LN1的数据结构:

|       | Age       | Tumor_Size | Ki67      | ER        | PR        | HER2      | \ |
|-------|-----------|------------|-----------|-----------|-----------|-----------|---|
| count | 11.000000 | 11.000000  | 11.000000 | 11.000000 | 11.000000 | 11.000000 |   |
| mean  | 57.090909 | 3.827273   | 45.454545 | 0.454545  | 0.181818  | 0.727273  |   |
| std   | 5.923758  | 0.718458   | 31.894713 | 0.522233  | 0.404520  | 0.467099  |   |
| min   | 49.000000 | 2.900000   | 10.000000 | 0.000000  | 0.000000  | 0.000000  |   |
| 25%   | 52.500000 | 3.350000   | 17.500000 | 0.000000  | 0.000000  | 0.500000  |   |
| 50%   | 56.000000 | 3.700000   | 40.000000 | 0.000000  | 0.000000  | 1.000000  |   |
| 75%   | 61.000000 | 4.250000   | 70.000000 | 1.000000  | 0.000000  | 1.000000  |   |
| max   | 67.000000 | 5.100000   | 95.000000 | 1.000000  | 1.000000  | 1.000000  |   |

| Lymph_Node |      |
|------------|------|
| count      | 11.0 |
| mean       | 1.0  |
| std        | 0.0  |
| min        | 1.0  |
| 25%        | 1.0  |
| 50%        | 1.0  |
| 75%        | 1.0  |
| max        | 1.0  |

-----连续变量-----

Age的正态分析:

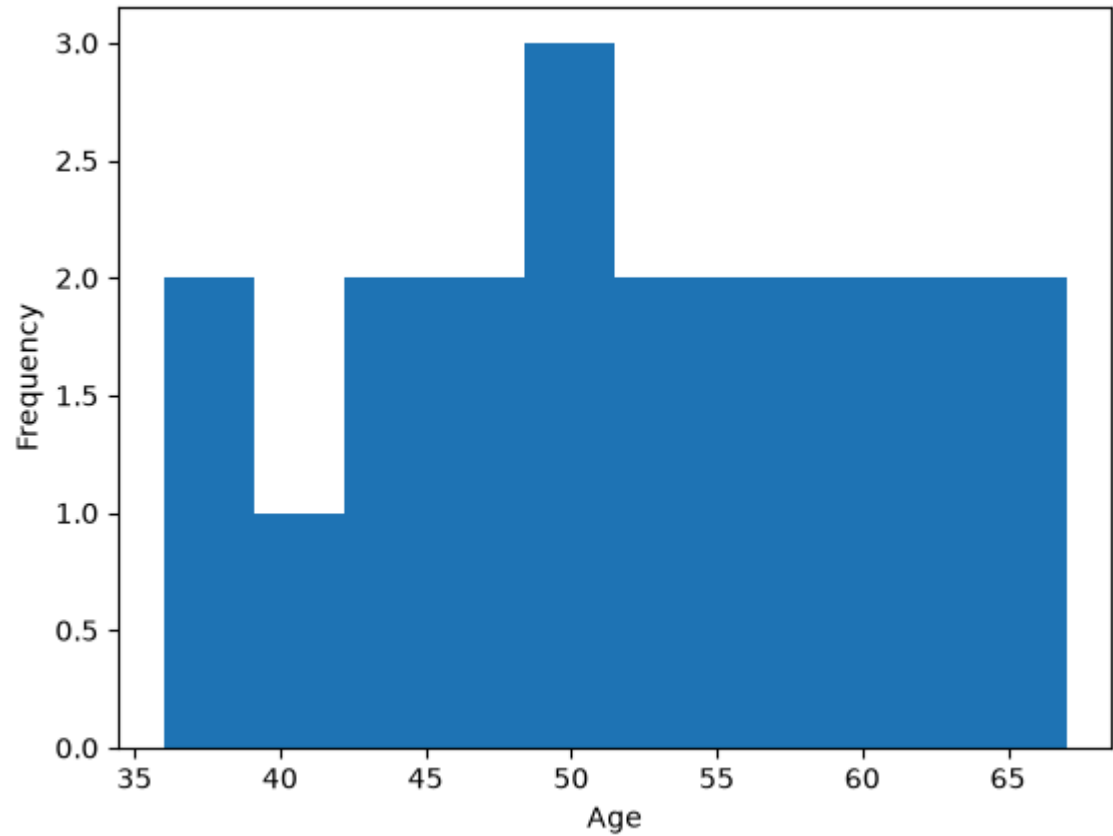
Age的p值: 0.9788032525063932

因为Age的p值大于0.05, 所以没有理由认为Age不符合正态分布, 因此下一步需要进行t检验。

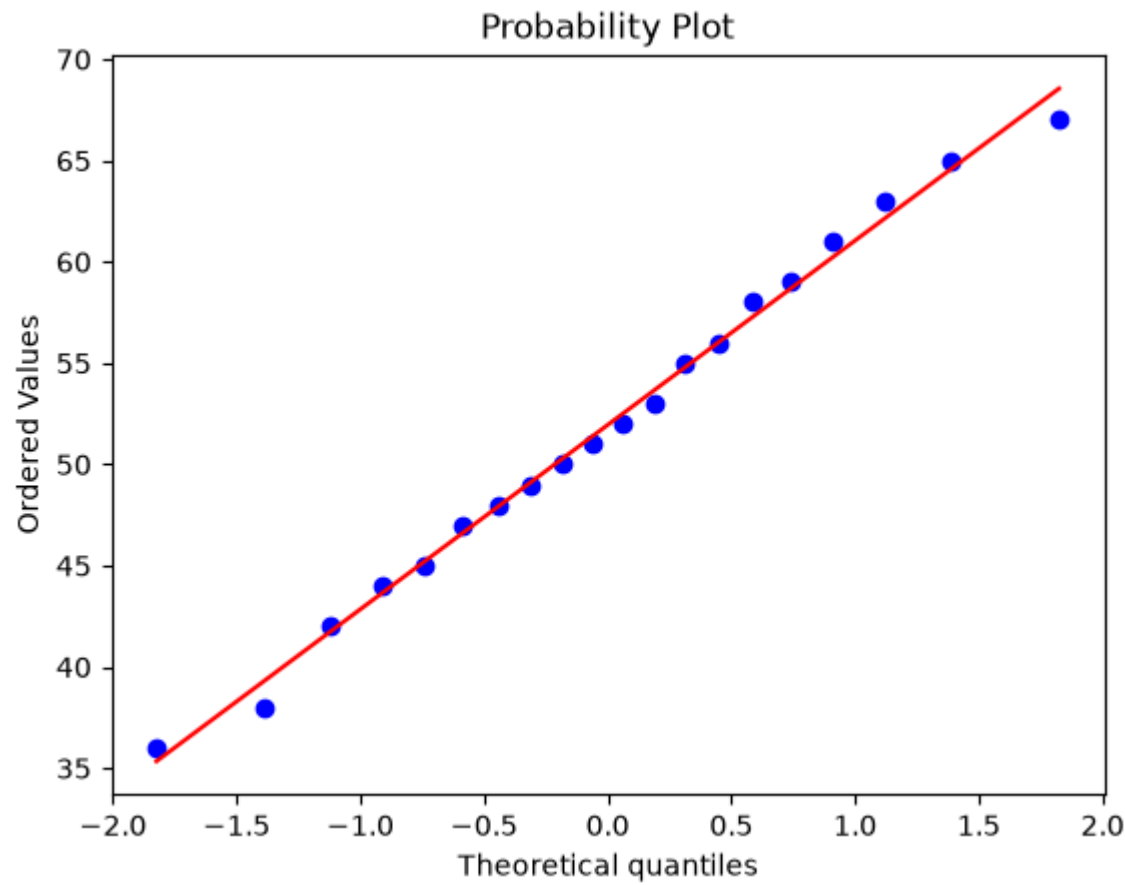
LN(-)Age的p值: 0.5600457012506541

LN(+)Age的p值: 0.6684407383707959

Age的直方图:



Age的Q-Q图:



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Tumor\_Size的正态分析:

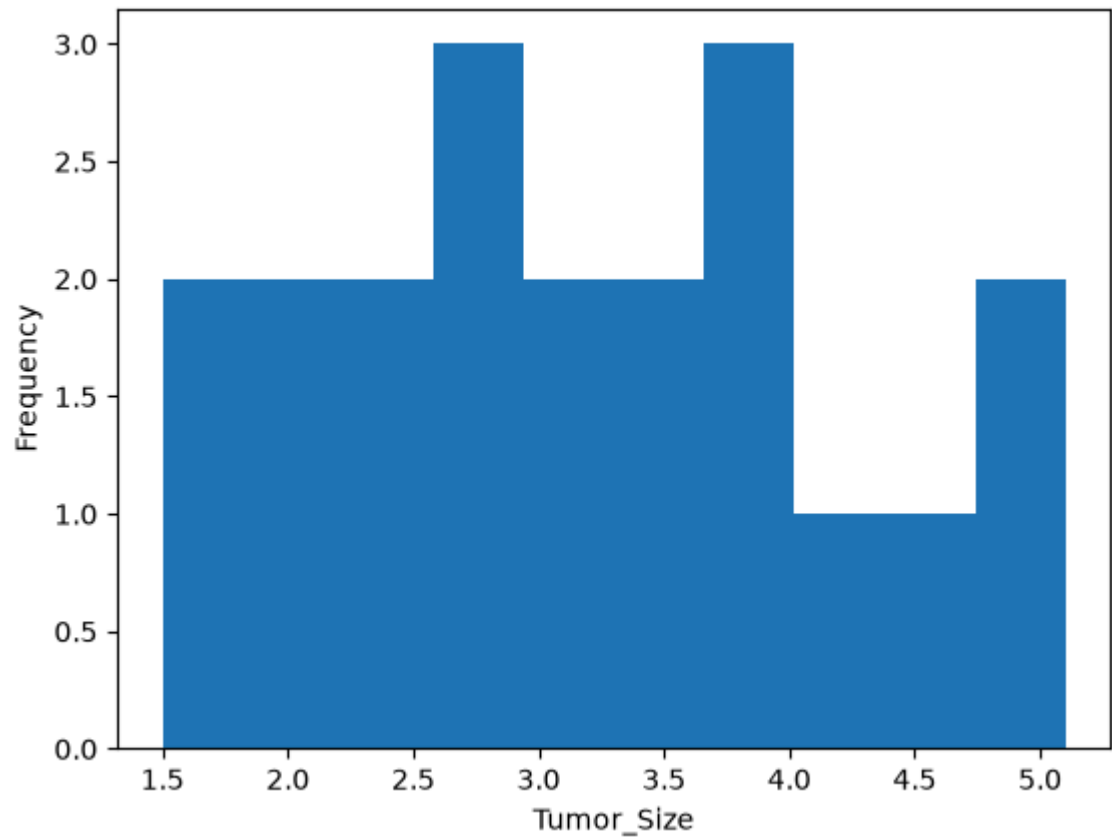
Tumor\_Size的p值: 0.9162922826518878

因为Tumor\_Size的p值大于0.05, 所以没有理由认为Tumor\_Size不符合正态分布, 因此下一步需要进行t检验。

LN(-)Tumor\_Size的p值: 0.15467787397132526

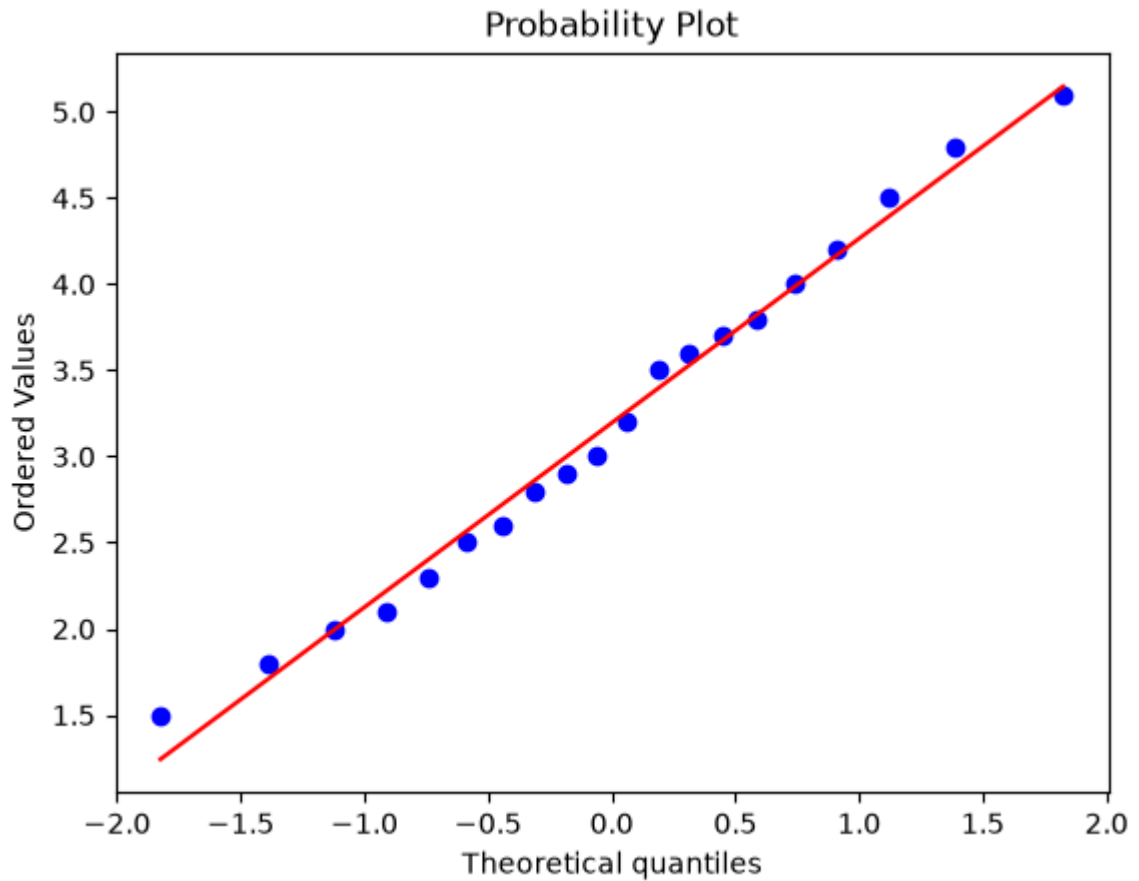
LN(+)Tumor\_Size的p值: 0.5763276869391712

Tumor\_Size的直方图:



Tumor\_Size的Q-Q图:



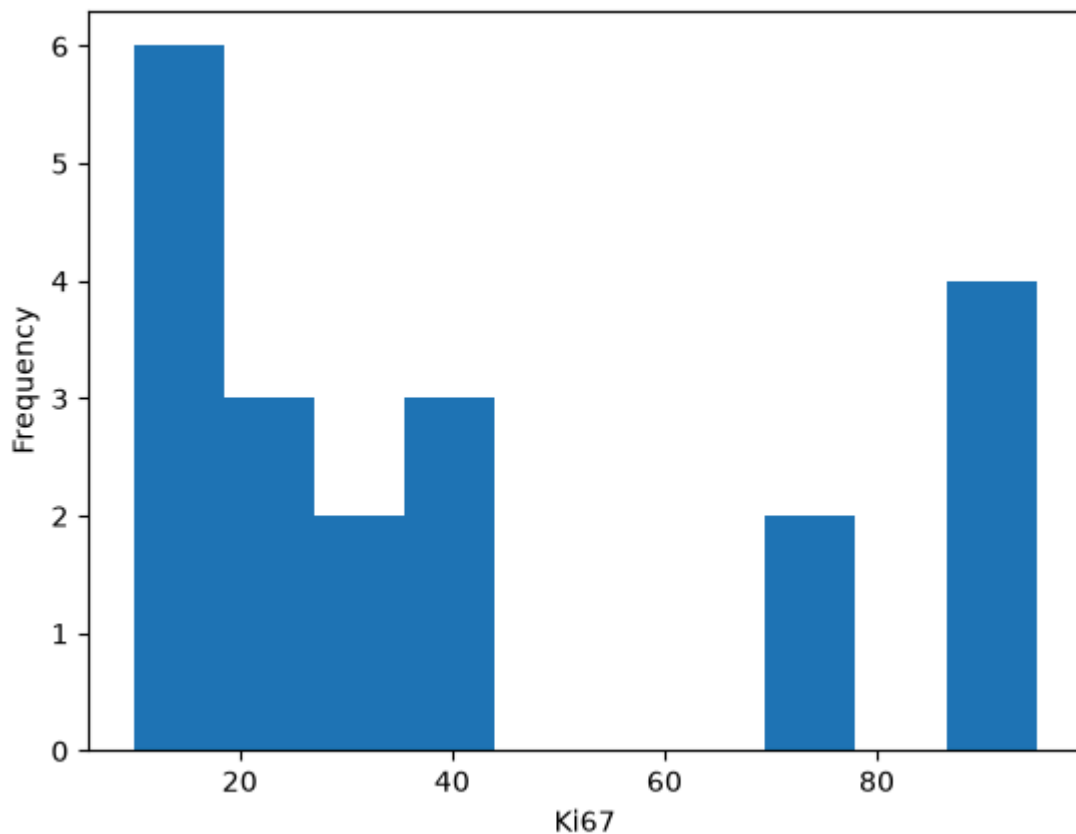


-----  
Ki67的正态分析:

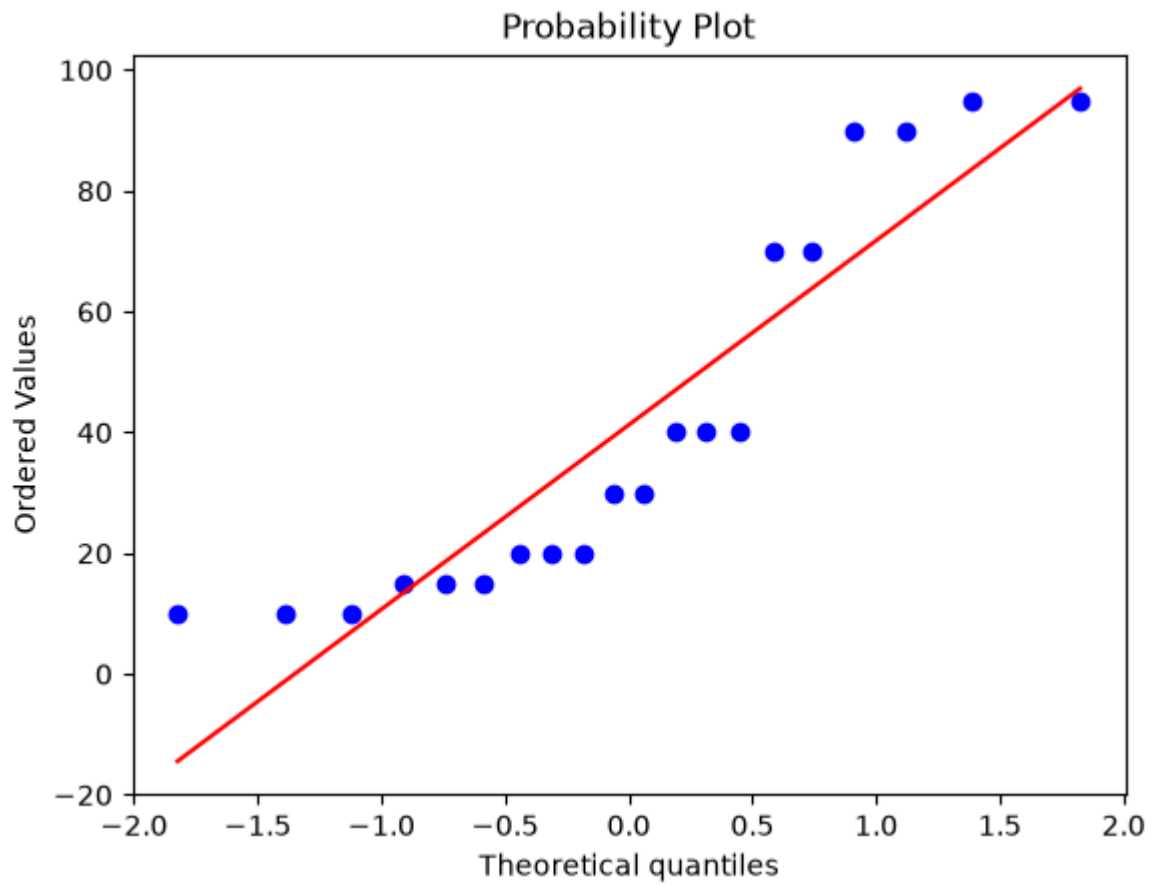
Ki67的p值: 0.002156689932776424

因为Ki67的p值小于0.05, 因此Ki67不符合正态分布, 因此下一步要进行Mann-Whitney U检验。

Ki67的直方图:



Ki67的Q-Q图:



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Age的t检验:
Age_ttest的p值: 0.0017755616146161615
因为Age_ttest的p值小于0.05, 所以两组的年龄差异有统计学意义。
-----
Tumor_Size的t检验:
Tumor_Size_ttest的p值: 0.0007047471803138009
因为Tumor_Size的p值小于0.05, 所以两组的肿瘤大小差异有统计学意义。
-----
Ki67的Mann Whitney U检验:
Ki67_U检验的p值: 0.42151799737413176
因为Ki67的p值大于0.05, 所以两组的Ki67差异有统计学意义。
-----
-----分类变量-----
ER与Lymph_Node的交叉表:
Lymph_Node      0      1
ER
0                5.0    30.0
1                40.0    25.0
PR与Lymph_Node的交叉表:
Lymph_Node      0      1
PR
0                10.0    45.0
1                35.0    10.0
HER2与Lymph_Node的交叉表:
Lymph_Node      0      1
HER2
0                35.0    15.0
1                10.0    40.0
-----
ER的chi square检验:
ER的p值: 1.562979606250842e-05
ER的expected: [[15.75 19.25]
 [29.25 35.75]]
因为样本量太小, 因此再进行一次fisher检验。
ER的fisher检验:
ER的p值: 5.937366666164522e-06
因为p值大于0.05, 所以ER和淋巴结转移之间没有统计学关联。
-----
PR的chi square检验:
PR的p值: 8.53304779969515e-09
PR的expected: [[24.75 30.25]
 [20.25 24.75]]
因为样本量太小, 因此再进行一次fisher检验。
PR的fisher检测:
PR的p值: 1.8621033784604504e-09
因为p值小于0.05, 所以PR与淋巴结转移之间有统计学关联。
-----
HER2的chi square:
HER2的p值: 1.4057935180978473e-06
HER2的expected: [[22.5 27.5]
 [22.5 27.5]]
因为样本量太小, 因此再进行一次fisher检验。
HER2的fisher检测:
HER2的p值: 8.356307654519182e-07
因为p值大于0.05, 所以HER2和淋巴结转移有统计学关联。
-----
-----Table1 Baseline Characteristics-----
Variable      Overall      LN(-)      LN(+)
0      Age  51.95±8.67  45.67±7.33  57.09±5.92
1  Tumor_Size  3.20±1.02  2.42±0.78  3.83±0.72
2      Ki67  30(15,70)  20(15,40)  45.45±31.89
3      ER      20      45%      55%

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4 PR 20 45% 55%

5 HER2 20 45% 55%

Baseline Characteristics Table1.xlsx保存成功

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-----Univariate Logistic Regression Analysis-----

Univariate Logistic Regression Analysis Table:

|   | Variables  | Uni_OR    | Uni_95%CI_low | Uni_95%CI_high | Uni_P    |
|---|------------|-----------|---------------|----------------|----------|
| 0 | Age        | 1.329203  | 1.036229      | 1.705010       | 0.025084 |
| 1 | Tumor_Size | 13.518047 | 1.534596      | 119.078654     | 0.018989 |
| 2 | Ki67       | 1.010267  | 0.980639      | 1.040791       | 0.501194 |
| 3 | ER         | 0.104167  | 0.009509      | 1.141126       | 0.064044 |
| 4 | PR         | 0.063492  | 0.007072      | 0.570034       | 0.013821 |
| 5 | HER2       | 9.333333  | 1.193455      | 72.990686      | 0.033296 |

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-----Multivariate Logistic Regression Analysis-----

Multivariate Logistic Regression Analysis Table:

|   | Variables  | Multi_OR     | Multi_95%CI_low | Multi_95%CI_high | Multi_P  |
|---|------------|--------------|-----------------|------------------|----------|
| 0 | Age        | 4.406498e-01 | 4.137368e-02    | 4.693135e+00     | 0.497150 |
| 1 | Tumor_Size | 1.515478e+06 | 7.026552e-05    | 3.268563e+16     | 0.241103 |
| 2 | Ki67       | 1.061867e+00 | 9.717194e-01    | 1.160378e+00     | 0.184782 |
| 3 | ER         | 3.605624e+04 | 6.178724e-05    | 2.104079e+13     | 0.308264 |
| 4 | PR         | 2.169745e-02 | 1.618976e-06    | 2.907884e+02     | 0.429511 |
| 5 | HER2       | 8.366226e-01 | 6.838726e-07    | 1.023491e+06     | 0.980101 |

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-----final\_table-----

|   | Variables  | Uni_OR    | Uni_95%CI_low | Uni_95%CI_high | Uni_P \  |
|---|------------|-----------|---------------|----------------|----------|
| 0 | Age        | 1.329203  | 1.036229      | 1.705010       | 0.025084 |
| 1 | Tumor_Size | 13.518047 | 1.534596      | 119.078654     | 0.018989 |
| 2 | Ki67       | 1.010267  | 0.980639      | 1.040791       | 0.501194 |
| 3 | ER         | 0.104167  | 0.009509      | 1.141126       | 0.064044 |
| 4 | PR         | 0.063492  | 0.007072      | 0.570034       | 0.013821 |
| 5 | HER2       | 9.333333  | 1.193455      | 72.990686      | 0.033296 |

|   | Multi_OR     | Multi_95%CI_low | Multi_95%CI_high | Multi_P \ |
|---|--------------|-----------------|------------------|-----------|
| 0 | 4.406498e-01 | 4.137368e-02    | 4.693135e+00     | 0.497150  |
| 1 | 1.515478e+06 | 7.026552e-05    | 3.268563e+16     | 0.241103  |
| 2 | 1.061867e+00 | 9.717194e-01    | 1.160378e+00     | 0.184782  |
| 3 | 3.605624e+04 | 6.178724e-05    | 2.104079e+13     | 0.308264  |
| 4 | 2.169745e-02 | 1.618976e-06    | 2.907884e+02     | 0.429511  |
| 5 | 8.366226e-01 | 6.838726e-07    | 1.023491e+06     | 0.980101  |

|   | Uni_OR_CI          | Multi_OR_CI                            |
|---|--------------------|----------------------------------------|
| 0 | 1.33(1.04-1.71)    | 0.44(0.04-4.69)                        |
| 1 | 13.52(1.53-119.08) | 1515477.65(0.0-3.2685627132824944e+16) |
| 2 | 1.01(0.98-1.04)    | 1.06(0.97-1.16)                        |
| 3 | 0.1(0.01-1.14)     | 36056.24(0.0-21040793759940.05)        |
| 4 | 0.06(0.01-0.57)    | 0.02(0.0-290.79)                       |
| 5 | 9.33(1.19-72.99)   | 0.84(0.0-1023490.85)                   |

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-----paper table-----

|   | Variables  | Uni_OR_CI          | Uni_P \  |
|---|------------|--------------------|----------|
| 0 | Age        | 1.33(1.04-1.71)    | 0.025084 |
| 1 | Tumor_Size | 13.52(1.53-119.08) | 0.018989 |
| 2 | Ki67       | 1.01(0.98-1.04)    | 0.501194 |
| 3 | ER         | 0.1(0.01-1.14)     | 0.064044 |
| 4 | PR         | 0.06(0.01-0.57)    | 0.013821 |
| 5 | HER2       | 9.33(1.19-72.99)   | 0.033296 |

|   | Multi_OR_CI     | Multi_P  |
|---|-----------------|----------|
| 0 | 0.44(0.04-4.69) | 0.497150 |

```
1 1515477.65(0.0-3.2685627132824944e+16) 0.241103
2          1.06(0.97-1.16) 0.184782
3      36056.24(0.0-21040793759940.05) 0.308264
4          0.02(0.0-290.79) 0.429511
5          0.84(0.0-1023490.85) 0.980101
paper_table.xlsx保存成功
```

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